

Jie Gao

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Malone Postdoctoral Fellow, Johns Hopkins University

Department of Computer Science

Malone Center for Engineering in Healthcare & Center for Language and Speech Processing

Malone Hall, 3400 N Charles St, Baltimore, MD 21218, USA

Research Interests

- Human-Centered Agentic AI; Human-Agent Collaboration; AI-assisted Qualitative Data Analysis; Trustworthy AI

My research focuses on human-AI collaboration for knowledge work: *as AI systems grow from single tools to autonomous multi-agent pipelines, how do we keep humans meaningfully in control?* I address this along two directions. **Bring AI to Humans:** Integrating LLM-based agents into human analytical workflows for qualitative data analysis, code comprehension, and subjective data annotation, ensuring that AI augments rather than replaces human judgment. **Bring Humans to AI:** Building oversight frameworks and evaluation methodology so that planning, memory, and tool use in multi-agent systems remain human-inspectable, with evaluation grounded in human judgment rather than automated metrics alone.

Employment Experience

Present– **Johns Hopkins University**, Baltimore, MD, USA

2025.03 **Malone Postdoctoral Fellow**, Department of Computer Science & Malone Center for Engineering in Healthcare & Center for Language and Speech Processing

Advisors: Mark Dredze, Ziang Xiao, Chien-Ming Huang

[Malone Postdoctoral Fellowship](#)

2025.02– **Singapore-MIT Alliance for Research and Technology (SMART)**, Singapore

2024.02 Postdoctoral Associate, Mens Manus Machina (M3S) team

Advisor: Thomas W. Malone

Education

2024.01– **Singapore University of Technology and Design**, Singapore

2019.09 Ph.D. in Information Systems Technology and Design

PhD Thesis: Tool Support for Human-AI Collaboration in Qualitative Data Analysis

[PDF](#)

Advisor: Simon Perrault

[SUTD PhD fellowship](#)

2019.06– **Zhejiang University**, Hangzhou, China

2017.09 M.Eng. in Software Engineering (Information Product Design)

[Direct admission based on outstanding undergraduate performance](#)

2017.06– **Shanghai University**, Shanghai, China

2013.09 B.Eng. in Digital Media Technology

[Top 1% of class](#)

Visiting Research Experience

- 2023.01– **University of Notre Dame**, Indiana, USA
2023.07 Visiting Ph.D. Student
Host Advisor: Toby Jia-jun Li
- 2022.06– **National University of Singapore**, Singapore
2022.12 Visiting Ph.D. Student
Host Advisor: Shengdong Zhao

Manuscript and Ongoing Work

- 1 Zhimin Lin, Kun Cheng, **Jie Gao***. **A Multi-Agent Framework for Qualitative Data Analysis.** *In submission to EMNLP2026* *Corresponding Author
- 2 **Jie Gao**, Mian Zhong, Jonathan Ivey, Chien-Ming Huang, Ziang Xiao, Mark Dredze. **Evaluation Framework for LLM-assisted Qualitative Data Analysis.** *In preparation to EMNLP2026*
- 3 **Jie Gao**, Thomas Malone, et al. **Not just Prompt Engineering! A Taxonomy for Human-LLM Interaction Modes.** *In submission to TOCHI2026* [manuscript](#) [Website](#)
- 4 **Jie Gao**, Zhiyao Shu, Shun Yi Yeo, Alok Prakash, Chien-Ming Huang, Mark Dredze, Ziang Xiao. **Efficiency with Rigor! A Trustworthy LLM-powered Workflow for Qualitative Data Analysis.** *In submission to TOCHI2026*, <https://arxiv.org/abs/2501.00775> [arXiv](#) [Tool Website](#)

Key Publications

- 1 **Jie Gao**, Yue Xue, Xiaofei Xie, SoeMin Thant, Erika Lee, Bowen Xu. **Understanding Codebase like a Professional! Human-AI Collaboration for Code Comprehension.** *The 34th IEEE/ACM International Conference on Program Comprehension (ICPC 2026)* [Cross-disciplinary collaboration with Software Engineering; Acceptance Rate: ~26.8%](#) [PDF](#) [Website](#)
- 2 **Jie Gao**, Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, Simon Perrault. **CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models.** *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)* [Highly cited in its domain; Citations: 180+, Jan, 2026; Acceptance Rate: ~23.6%](#) [PDF](#) [Website](#)
- 3 **Jie Gao**, Kenny Tsu Wei Choo, Junming Cao, Roy Ka Wei Lee, Simon Perrault. **CoAICoder: Examining the Effectiveness of AI-assisted Human-to-Human Collaboration in Qualitative Analysis.** *ACM Transactions on Computer-Human Interaction (TOCHI 2023)* [Highly cited in its domain; Citations: 100, Jan, 2026; Peer-reviewed journal \(impact factor 3.5+\)](#) [PDF](#)
- 4 **Jie Gao***, Simret Araya Gebreegziabher*, Kenny Tsu Wei Choo, Toby Jia-Jun Li, Simon Tangi Perrault, Thomas W Malone. **A Taxonomy for Human-LLM Interaction Modes: An Initial Exploration.** *Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI EA 2024)* [Highly cited in its domain; Citations: 80+, Jan, 2026](#) [PDF](#)

- 5 **Jie Gao**, Xiayin Ying, Junming Cao, Yifan Yang, Pin Sym Foong, Simon Perrault. **Differences of Challenges of Working from Home (WFH) between China and the US during COVID-19.** *Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI EA 2022)* [PDF](#)
- 6 Zheng Zhang, **Jie Gao**, Ranjodh Singh Dhaliwal, Toby Jia-Jun Li. **VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping.** *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2023)* [Highly cited in its domain; Citations: 170+, Jan, 2026](#) [PDF](#)
- 7 Nuwan Janaka, **Jie Gao**, Lin Zhu, Shengdong Zhao, Lan Lyu, Peisen Xu, Maximilian Nabokow, Silang Wang, Yanch Ong. **GlassMessaging: Towards Ubiquitous Messaging Using OHMDs.** *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT 2023)* [PDF](#)
- 8 ShunYi Yeo, Gionnieve Lim, **Jie Gao**, Weiyu Zhang, Simon Tangi Perrault. **Help Me Reflect: Leveraging Self-Reflection Interface Nudges to Enhance Deliberativeness on Online Deliberation Platforms.** *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)* [PDF](#)
- 9 Junming Cao, Bihuan Chen, Longjie Hu, **Jie Gao**, Kaifeng Huang, Xuezhi Song, Xin Peng. **Understanding Complexity and Its Impact on Testing in ML-Enabled Systems.** *IEEE International Conference on Software Maintenance and Evolution (ICSME 2023)* [PDF](#)

Peer-Reviewed Short Papers

- 1 **Jie Gao**, Yuchen Guo, Toby Jia-jun Li, Simon Perrault. **CollabCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis.** *ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW Demo 2023)*
- 2 Ruyuan Wan, Haonan Wang, Ting-Hao ‘Kenneth’ Huang, **Jie Gao***. **From Noise to Nuance: Enriching Subjective Data Annotation through Qualitative Analysis.** *The 4th HCI+NLP Workshop at the Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)* [Advising Author](#)
- 3 **Jie Gao**, Leijing Zhou, Miaomiao Dong, Fan Zhang. **Expressive Plant: A Multisensory Interactive System for Sensory Training of Children with Autism.** *ACM International Joint Conference on Pervasive and Ubiquitous Computing (Ubicomp Poster 2018)*
- 4 Marianne Aubin Le Quéré, Hope Schroeder, Casey Randazzo, **Jie Gao**, Ziv Epstein, Simon Perrault, David Mimno, Louise Barkhuus, Hanlin Li. **LLMs as Research Tools: Applications and Evaluations in HCI Data Work.** *Extended Abstract of the ACM Conference on Human Factors in Computing Systems (CHI2024 Workshop Proposal)* [PDF](#)

Honors and Awards

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|------|---|
| 2024 | Malone Postdoctoral Fellowship
Johns Hopkins University |
| 2019 | SUTD PhD Fellowship
Singapore University of Technology and Design |
| 2017 | Excellent Graduate Student
Zhejiang University |

2015 **Excellent Undergraduate Student**
Shanghai University

Invited Talks & Presentations

- 2025.11 **Human-AI Collaboration for Complex Unstructured Data Analysis**
Western Michigan University
- 2025.10 **From Automation to Human-AI Collaboration: Methods and Applications for Insights Extraction in Unstructured Data**
Malone Center Research Lunch at Johns Hopkins University
- 2025.07 **AI for Text Production and Analysis**
HCIX Research Club
- 2025.04 **Designing Human-AI Teams to Extract Insights from Unstructured Data—Methods and Applications**
HCI Seminar at Johns Hopkins University
- 2025.03 **Human-Centric Software Engineering in the Era of Large Language Models**
Singapore Management University
- 2025.02 **Extracting Insights from (Un)structured Data: Methods and Applications**
M3S Seminar in Mens Manus Machina (M3S) team at Singapore-MIT Alliance for Research and Technology

Academic Service

Associate Chair, ACM Conference on Human Factors in Computing Systems (CHI 2025 & 2026), Main Track

Reviewer, Annual ACM Symposium on User Interface Software and Technology (UIST 2025)

Reviewer, Computers in Human Behavior Reports 2025

Reviewer, International Journal of Human-Computer Interaction (IJHCI 2025, 2026)

Reviewer, ACM Transactions on Interactive Intelligent Systems (TiiS 2024, 2025)

Associate Chair, ACM Conference on Human Factors in Computing Systems (CHI 2022, 2023), Late-Breaking Work Track

Reviewer, ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2022, 2023)

Reviewer, ACM Conference on Human Factors in Computing Systems (CHI 2023, 2024)

Reviewer, International Conference on Artificial Intelligence in Education (AIED 2023)

Mentoring Experience

Spring **Yifan Zhang (PhD Student)**
2026 National University of Singapore

Fall **Mian Zhong (PhD Student)**
2025 Johns Hopkins University

Fall **Jonathan Ivey (PhD Student)**
2025 Johns Hopkins University

Summer- **Haonan Wang (Master)**
Fall Johns Hopkins University
2025

Summer **Ruyuan Wan (PhD Student)**
2024 Pennsylvania State University

Summer- **Zhiyao Shu (Master, Remote Research Intern)**
Spring University of California, Berkeley (Graduated). Now PhD student at George Mason University
2024–2025

Summer **Xiaoxian Zhang (Master, Research Intern)**
2024 National University of Singapore

Summer **Erika Lee (Undergraduate, Research Intern)**
2024 University of California, San Diego

Fall **Yuchen Guo (Undergraduate, Research Intern)**
2022 Singapore University of Technology and Design

Fall **Tianqin Zhang (Undergraduate, Research Intern)**
2022 Singapore University of Technology and Design

Teaching Experience

Fall **Teaching Assistant, 50.002: Introduction to Algorithms (Singapore University of Technology and Design)**
2020

Summer **Teaching Assistant, 50.017: Graphics and Visualization (Singapore University of Technology and Design)**
2020

References (Ranked by Priority)

Mark Dredze (mdredze@cs.jhu.edu, +1 (410) 516-6786)
Professor, Johns Hopkins University

Thomas W. Malone (malone@mit.edu, +1 (617) 253-6843)
Professor, Massachusetts Institute of Technology

Simon Perrault (simon.perrault@telecom-paris.fr, +33 664871217)
Professor, Télécom Paris, Institut Polytechnique de Paris

Ziang Xiao (ziang.xiao@jhu.edu, +1 (217)281-2005)
Assistant Professor, Johns Hopkins University

Toby Jia-Jun Li (toby.j.li@nd.edu, +1 (574) 631-5375)
Assistant Professor, University of Notre Dame